Solar ESS Co. Ltd.
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ELECTRICIAN JOB SHEET · PROJECT BASED

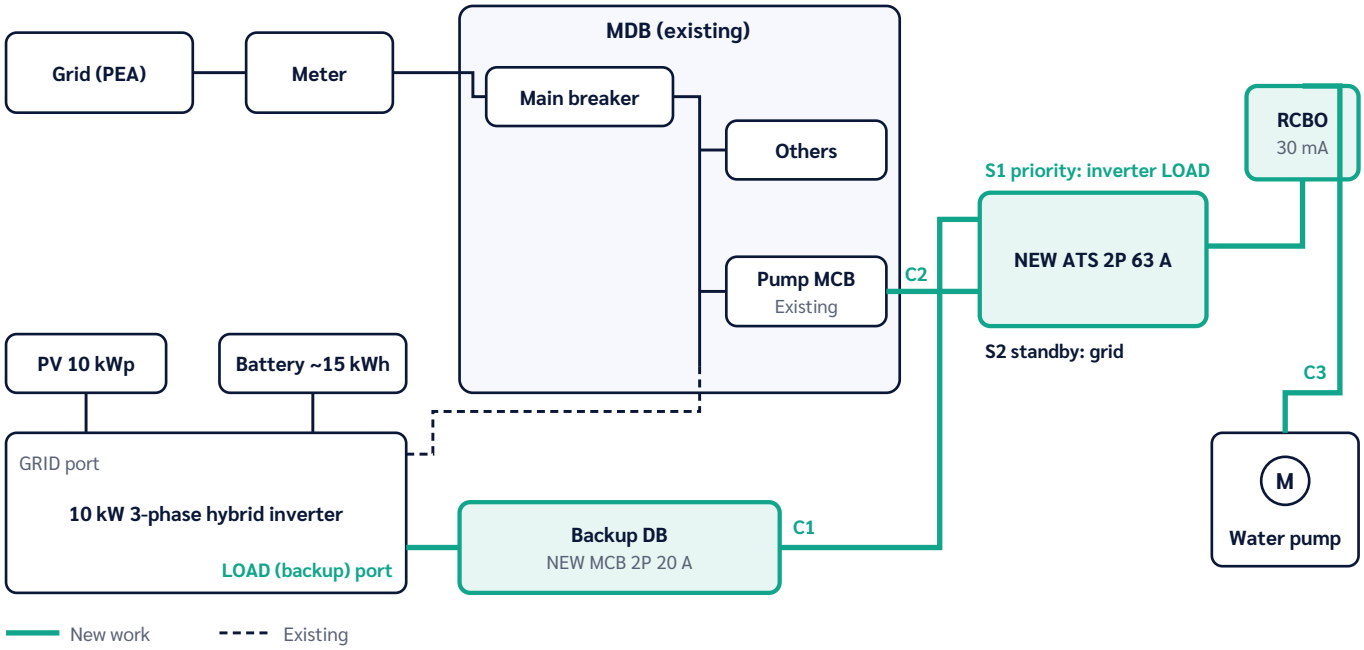
Put the water pump on battery backup and install an ATS at the MDB

Harmony@Huailan Ecolodge, Chiang Mai · Job no. JOB-HHL-PUMP-01 · Issued by Solar ESS Working Group · 2 Nov 2026

The job in short

- Today the water pump is fed from the grid through the MDB. When the grid goes off, the pump stops.
- This job: run a new circuit from the inverter's **LOAD (backup) port** to a **new ATS at the MDB**, and from the ATS to the water pump.
- **Normal:** the pump is fed through the inverter (S1). **Blackout:** the pump keeps running from the battery automatically. **Inverter off, faulty or under service:** the ATS transfers the pump to the grid (S2).
- Never connect the LOAD port in parallel with the grid. The ATS must switch both L and N, with a mechanical interlock (break-before-make).

Single-line diagram



C1 new cable from the backup DB (LOAD port) to ATS input S1 · C2 cable from the existing pump MCB in the MDB to ATS input S2 · C3 cable from the ATS output through the RCBO to the pump. Shown for a single-phase pump; for a three-phase pump use a 4P ATS, 4P MCB and 5-core cable.

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Existing system on site

Item	Details
Solar panels	10 kWp
Inverter	10 kW three-phase low-voltage hybrid · LOAD (backup) port max 10 kW total / 5 kW per phase · surge 2× for 10 s
Battery	Lithium 48 V 300 Ah (~15 kWh) · set to discharge down to 25%
MDB	Steel MDB with volt/ammeters; feeders: Residence, Pavilion, Guest House, Outside Toilet and spares
Backup load today	Only 30–130 W (almost every circuit is on the grid side)

Scope of work

1. **Survey first:** record the pump nameplate (kW, V, 1- or 3-phase, starting method), measure running and starting current, find the pump circuit in the MDB, measure cable runs C1, C2, C3 and find a free LOAD-port way. Send the data to Solar ESS to confirm before buying materials.
2. **Isolate safely (LOTO):** switch off the MDB main breaker, the inverter GRID and LOAD AC breakers, the inverter, the battery breaker and the PV DC isolators. Prove dead at every point before touching any conductor.
3. **Backup DB:** install a new MCB 2P 20 A fed from the inverter LOAD port, on the least-loaded phase (L1 recommended).
4. **Cable C1:** THW 3×4 mm² (L, N, PE) in ¾" IMC/EMT conduit from the new MCB to ATS input S1. Use 6 mm² if the run is over 30 m (voltage drop ≤ 3%).
5. **ATS:** install a 2P 63 A ATS in an IP54 enclosure beside the MDB (or inside the MDB if there is room). Set S1 (inverter) as priority, transfer delay to S2 about 5 s, return to S1 after 30–60 s.
6. **Cable C2:** move the outgoing cable of the existing pump MCB in the MDB to ATS input S2 (check the MCB rating suits the pump).
7. **Cable C3 and RCBO:** from the ATS output through a 1P+N 20 A 30 mA RCBO to the pump, in IMC/EMT (waterproof conduit and glands outdoors). Earth the pump.
8. **Neutral:** never join the LOAD-port neutral to the MDB neutral bar; it must pass through the ATS only. Protective earth (PE) may be common.
9. **Labels:** on the ATS, the new MCB and the MDB: "Water pump fed from the inverter – live from the battery during a grid outage".
10. **Test and hand over** per the test table below, with photos, an as-built diagram and the Solar ESS report form.

> **Danger:** the inverter LOAD port stays live from the battery and the solar panels even when the grid is off. Always switch off the inverter, the battery and the DC isolators first, lock out and tag out (LOTO), and prove dead before work. Tell the lodge owner in advance: the water pump will be off during the work.

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Materials (minimum specification)

Premium materials only. Give the brand and size of every item in your quote.

#	Item	Minimum specification	Qty
1	ATS (automatic transfer switch)	2P 63 A 230 VAC, with controller, mechanical interlock, break-before-make, adjustable delays	1
2	ATS enclosure	Steel or plastic IP54, with DIN rail and terminals (if the ATS cannot fit in the MDB)	1
3	MCB, inverter side	2P 20 A curve C, 6 kA (to suit the actual pump)	1
4	RCBO, pump side	1P+N 20 A 30 mA Type A	1
5	Cable	THW 4 mm² copper (black, blue, green/yellow), length as measured; 6 mm² if over 30 m	As measured
6	Conduit	¾" IMC or EMT with fittings, saddles, pull boxes and a waterproof flexible connection at the pump	As measured
7	Consumables	Lugs, ferrules, cable markers, warning labels, cable ties, anchors	1 set

Tests (record every reading)

#	Test	Pass when
1	Insulation of cables C1, C2, C3 at 500 VDC (L–N, L–PE, N–PE)	≥ 1 MΩ
2	Polarity and earth continuity to the pump body	Correct, PE continuous
3	Normal operation: ATS on S1, pump running through the inverter; record voltage at the pump, starting and running current	220–240 V, pump normal
4	Simulated blackout: open the MDB main breaker. The pump must keep running from the battery; run 10 min and restart 3 times	No stop, no inverter trip
5	Simulated inverter failure: open the new inverter-side MCB. The ATS must transfer to S2 and return to S1 when closed	Auto transfer and return
6	RCBO test button	Trips at once
7	Inverter display / monitoring app shows the pump power as backup load	Pump power seen on LOAD

To confirm on site (before the final price)

- Pump: kW, 1- or 3-phase, starting method and starting current. If above 2.2 kW or with a very high starting current, tell Solar ESS first: a soft starter may be needed.
- Cable run lengths C1, C2, C3 and conduit route.
- Free space in the MDB and in the backup DB; free LOAD-port terminals on the inverter.
- Existing terminal block (labels RM1, SM1, TM1, RKM, SKM, TKM, LLM, N): identify where every conductor goes before touching it.

How to quote

- Quote **labour** and **materials separately**, with the brand and size of every material.
- Pay guide for labour: **฿3,500** for 2 people, about 1 day including tests; materials and travel not included.
- Attach your electrician certificates and photos of similar work. The work must be done by a qualified electrician.
- Do not change any inverter settings; Solar ESS will set the inverter remotely if needed.
- Send your quote with the form on www.solaress.org/job.html.

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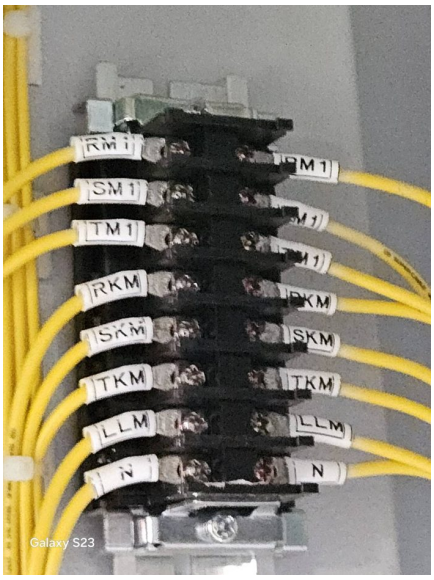
Site photos



Existing MDB (the ATS goes here)



Inverter room with AC and PV boxes



Existing terminal block (identify before work)

Qualifications and certificates of the Solar ESS team are furnished upon request or upon completion of contract.

Solar ESS Working Group